

| Question | Marking Guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Mark  | Comments                                                                                                                                                                                                                                                                                                      |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5.1      | 1. RNA (as genetic material);<br>2. Reverse transcriptase;<br>3. (Protein) capsomeres/capsid;<br>4. (Phospho)lipid (viral) envelope<br><b>OR</b><br>Envelope made of membrane;<br>5. Attachment proteins;                                                                                                                                                                                                                                                                                  | 4 max | Accept a labelled diagram.<br>1. Reject nucleus/DNA/plasmids.<br>3. Reject capsule.<br>4. Reject if HIV has a cell membrane or a cell wall.<br>5. Accept gp41 and/or gp 120.<br>5. Accept glycoprotein.<br>5. Accept description of attachment protein.<br>5. Ignore 'receptor protein'.<br>Ignore cytoplasm. |
| 5.2      | Automarked q – <input checked="" type="checkbox"/> 106                                                                                                                                                                                                                                                                                                                                                                                                                                     | 1     |                                                                                                                                                                                                                                                                                                               |
| 5.3      | 1. (All) have more T helper/CD4 cells;<br>2. <u>Lower</u> viral load <b>to</b> infect/destroy helper T/CD4 cells;<br>3. (So more/continued) activation of B cells/cytotoxic T cells/phagocytes;<br>4. (With B cells more/continued) production of plasma cells/antibodies<br><b>OR</b><br>(With cytotoxic T cells more/continued) ability to kill virus infected cells;<br>5. (More able to) destroy other microbes/pathogens<br><b>OR</b><br>(More able to) destroy mutated/cancer cells; | 3 max | 1. Accept <u>higher</u> proportion of T helper/CD4 to virus particles.<br>1. and 2. Statement must be comparative.<br>2. For 'infect' accept 'HIV does not reproduce in'.<br>3. Accept 'stimulation' for 'activation'.<br>4. Ignore reference to B cells acting as phagocytes/antigen-presenting cells.       |

